Data Management Plan

SQRT (Sub-zero Quantitative Research Team

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Key Reference Documents

1. UCD-DRD-2026-1
2. SQT-PDR-SEP-0
3. SQT-CDR-SDF-0

1 Executive Summary

This document will cover the data management strategy of the SQRT mission. This will include the delineation of data products to be transmitted to ground during the mission from those that will be stored onboard. Every product and its delivery method will be justified with reference to the mission requirements and constraints.

The data produced by this mission will be classified on their contribution to the mission outcome i.e. whether the data product is linked to a mission requirement (level 1), or technical requirement (level 2). This hierarchy will guide the identification of the data products to be transmitted to the ground segment, and those that will be stored onboard. It is imperative that sufficient data is downlinked to satisfy the defined mission goals, in the event that the TupperSat cannot be physically recovered. As will be detailed in sections 3 and 4, constraints on the data packet will inhibit the transmission of data. Regarding downlinked data, the rate of data transmission will be evaluated, verifying that it is congruent with mission requirements. The format of the data packet will also be outlined. Trade-offs will be required, and these will be outlined and justified.

The thermal sensor (Adafruit MLX90640) is a critical data source, lending importance to its calibration and giving it priority for storage. This document will be concerned with the management of raw figures and calibration constants returned by the device. Trade-off analysis will be necessary to ensure that a sufficient data rate is maintained during the calibration process. The process of interfacing with this component will be covered in greater detail in SQT:CDR:SDF:0.

The latter sections of this document will be concerned with the post-flight data formatting. An overview of the steps to be taken to analyse the mission data will be provided. This will include any processes that will need to be carried out to supply each team member with a common, interpretable data pack that can be analysed further individually for the post-mission report. A successful post-flight data processing regime will be essential to providing the grounds for a successful mission outcome.

2 Mission Overview

2.1 Outline

Mission Statement: The SQRT mission will produce and observe the flash evaporation of water via exposure to a low pressure and low temperature environment to investigate how such plumes thermally evolve.

This investigation into the thermal mechanisms behind cryovolcanic plumes will be done using a high-altitude demonstration. This will involve the sudden exposure of liquid water within the TupperSat to a colder, lower-pressure environment which will be generated by elevating the TupperSat system to an altitude between 20-25 km. At 23 km altitude, the average atmospheric pressure is about 34 mbar (approx. 3.4% of that at sea level) with a corresponding boiling point of water of approx. $27 - 28°C$. To perform this exposure, an opening mechanism, consisting of a valve, a trigger sensor (based on ambient pressure), and a servo motor, will allow a small volume of water to "see" a colder, less pressured environment within the confines of the TupperSat.

2.2 Mission Critical Data

A successful mission outcome will be measured against a number of predefined metrics. These will include the confirmation of exposure mechanism function, and the acquiring of readable thermal data exhibiting the subsequent flash evaporation process.

The mission statement above dictates the development of the mission and technical requirements found in UCD:CDR:SEP:0. In order to satisfy these requirements, and by extension, the mission objective, data will be required to verify the exposure of the water, and to quantify the thermal evolution displayed by the subsequent flash evaporation and cooling. The acquisition and storage of this information is imperative to maximising mission returns. Mission-critical data includes the thermal sensor data from within the low-pressure chamber, a failsafe thermometer at the valve opening to confirm exposure, and a further back up signal to confirm that the servo motor was activated i.e. valve was released.

3 Onboard Data Products

The onboard data products will be stored on a 32 GB SD card. The data products below will be stored onboard rather than downlinked to conform to requirements involving the data packet size and rate. Technical requirements UCD:TR:00120 & UCD:TR:00130 limit the total packet rate to 3 per minute (2 telemetry, 1 data).

3.1 Sources

Source	**Product Name**	**Data Rate**
Thermal Sensor	Adafruit MLX90640	98.304 kB/s (64 fps), 49.152 kB/s (32 fps)
Thermometer (Valve Proximity)	DS18B20	40 B/s
GPS	ANT-GPS-P20-SMA	200 B/s
Pressure Sensor (External)	MS5611-01BA03	10 B/s
Total (6 Hours)		**2.13 GB (64fps), 1.07 GB (32 fps)**

Table 1: Sources of Onboard data

The thermal sensor contains a number of internal registers through which the device performance can be customised. For instance, control register 1 - address 0x800D allows the user to define the IR refresh rate. Table 1 displays the data rate for both the 64 Hz and 32 Hz modes. Another control that can be adjusted is the resolution of the ADC. This could be useful when performing data compression for transmission [1].

From the table above, the total amount of data stored on board the SD card is expected to be approx. 2 GB. This is well below the capacity of the SD card, suggesting that the pressure and temperature sampling rates can be scaled up considerably.

4 Downlinked Data Products

Downlinked data products will consist of scaled down subsets of the data collected throughout the mission campaign. Science data packets will be transmitted once per minute, and will be required to contain sufficient information to satisfy the mission objectives in the event of non-recovery (as per requirement UCD:MR:0060).

4.1 Sources

Source	**Product Name**	**Data Transmission Rate**
Thermal Sensor	Adafruit MLX90640	96 bytes/minute
GPS	ANT-GPS-P20-SMA	16 bytes/minute
Pressure Sensor (External)	MS5611-01BA03	10 bytes/minute
Thermometer (Valve Proximity)	DS18B20	40 bytes/minute
Servo Motor Flag	DFROBOT	2 bytes/minute
Per Packet		**164 bytes/data packet**

Table 2: Sources of Transmitted Data

The total data rate per packet conforms with the allocated packet size of 256 bytes. The sources above and their respective rates of transmission are required to fulfill a number of the technical requirements found in SQT-CDR-SEP-0:

Thermal Sensor:
The initial plan was for all thermal sensor data to be stored onboard. However, in order to have sufficient data to confirm mission success in the event of non-recovery, it was decided that limited thermal sensor data would have to be transmitted. The entire array (32x24) could not be transmitted, so reduction methods would be necessary. One such method is downsampling, whereby the 768 pixel array is broken down into 4x4 frames. By averaging each frame, a new 8x6 array is created. The data rate has been reduced by a factor of 16 using this method. Alternative approaches include focusing on a particular region of interest to reduce the number of points being transmitted.

GPS:
The GPS data will be required to provide continuous longitude, latitude and altitude information to the ground segment. This data will allow for both the monitoring of mission processes (valve activation at suitable altitude) and the tracking of the TupperSat during descent. In accordance with requirements, the 3-dimensional (lat, lon, alt) location of the TupperSat will be transmitted at least once a minute. The GPS information stored will be in the form of a dictionary with keys: *Timestamp, Altitude, Latitude, Longitude*. The GPS information transmitted will take the form of 3 floats, corresponding to the latitude, longitude and altitude.

Servo Motor Flag:
This flag will activate when the servo motor has received the trigger activation signal and has released the valve. The flag will be transmitted with every data packet to ensure that confirmation of the valve's activation reaches the ground segment. Transmission of this flag will partially satisfy the mission objectives by confirming that the exposure mechanism has functioned. This flag will be paired with the measurements from the internal thermometer to confirm an increase in temperature due to exposed water movement.

4.2 Data Packet Format

The data packet formats for telemetry and science data transmissions are defined below. These formats are designed with the aim of minimising packet size, while communicating all necessary data.

4.2.1 Telemetry

The contents of the housekeeping/telemetry data packet will be dictated primarily by the customer's communications requirements. These requirements focus on accurately defining the TupperSat's location and monitoring the ambient conditions. At least two of these packets are required to be transmitted per minute. This is to facilitate continuous tracking of the TupperSat's location, a process that will mitigate the risk of non-recovery.

Information Required:
1. Timestamp

2. Altitude

3. Latitude/Longitude

4. Internal, External Temperatures

5. Atmospheric Pressure

PACKET IDENTIFICATION			TELEMETRY						
Packet Type	TupperSat ID	Packet #	Time (UTC)	Altitude (m)	Latitude	Longitude	Pressure	Temperature (E)	Temperature (I)
ASCII	ASCII	Binary	hhmmss	float	signed float	signed float	float	signed float	signed float
1 byte	3 bytes	2 bytes	3 bytes	4 bytes	4 bytes	4 bytes	4 bytes	4 bytes	4 bytes

Figure 1: Housekeeping Data Packet Format

Example:
T|SQT|0123|123456|23081.56|+54.12345|-004.67890|-040.562|028.973|0040.11

The proposed data packetization format is visualised above. The housekeeping packet will be comprised of two parts: one to identify the packet as being a telemetry packet belonging to the SQRT mission, another to report the timestamped telemetry. The data types for each field are also specified. Based on the Figure 1 above, the size of the downlinked housekeeping packet is expected to be 33 bytes, well below the 256 byte limit.

4.2.2 Data

The data packet will be designed to return all data pertaining to the experimental payload. This will include outputs from both the thermal sensor and the exposure mechanism.

Information Required:
1. Timestamp

2. Thermal Data

3. Pressure Trigger Status

4. Servo Motor Activation

PACKET IDENTIFICATION			DATA					
Packet Type	TupperSat ID	Packet #	Time (UTC)	Thermal Array	Servo Motor Flag	Trigger	Trigger Type	Valve Temp Rise
ASCII	ASCII	Binary	hhmmss	48 x int16	Binary	Binary	ASCII	Binary
1 byte	3 bytes	2 bytes	3 bytes	96 bytes	2 bytes	2 bytes	1 byte	2 bytes

Figure 2: Science Data Packet Format

Example:
D|SQT|0089|140204|[1941,1945,...,2022]|1|G|0

The downlinked thermal array data will be a downsampled version of the full 768 pixel array, producing a reduced array of 48 integer values per sample (in centicelsius). This reduced array acts as the secondary mission validation criterion, in case the TupperSat cannot be recovered. The array will take the form of a list of 48 integer values.

Additional science data to be downlinked will include the two validation flags, reported every minute. The first flag is the pressure trigger, indicating whether the triggering criteria have been satisfied. The trigger type field refers to the criteria used to activate the trigger and commence the exposure process. There are 3 tiers of triggers: the Good (Pressure), the Bad (Altitude) and the Ugly (Falling check). The ASCII character sent down in the data packet will inform the team of the trigger that was used (G,B or U). Following trigger activation, the next check will be used to confirm if the servo motor has received instruction to release the valve and has done so. Thermometer readings near the valve will be consulted to validate these flags and to verify that the water has been released i.e. that the temperature has increased. These checks will serve as tertiary mission validation criteria in the event of thermal array failure.

5 Data Analysis

The data analysis section of the mission will consist of the processing of retrieved data to yield interpretable results that can be used to demonstrate the success of the mission and perform scientific analysis of the physical processes at play.

This process will consist of:

1. Retrieval of data via recovery of onboard and transmitted data.
2. Storage of raw dataset for reference.
3. Identification and removal of any corrupted sentences/files.
4. Decoding of raw files based on data packet format outlined in section 4.

The flowchart below illustrates the approach that will be adopted in the post-mission data analysis campaign.

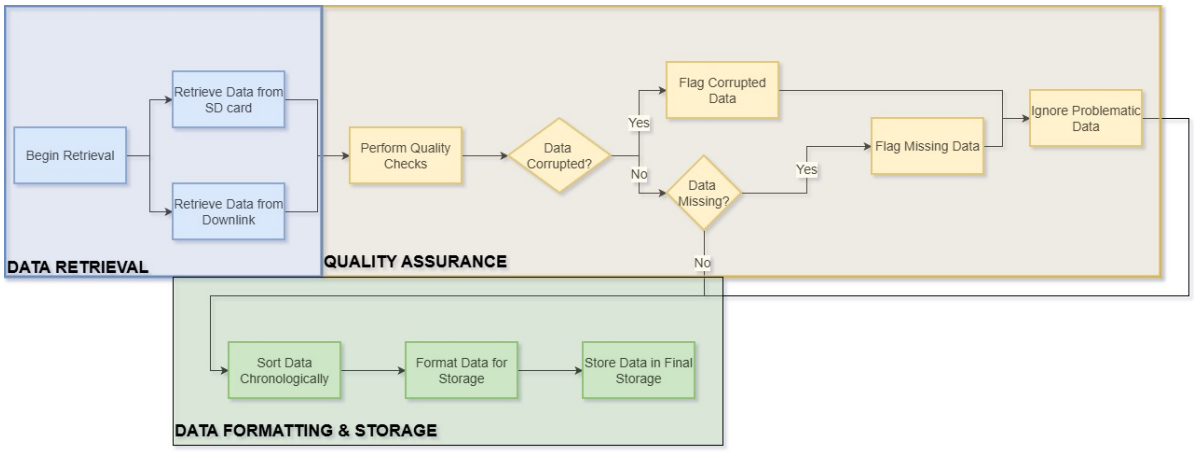


Figure 3: Data Analysis Flowchart

The data originating from each sensor (thermometer, barometer or thermal sensor) will vary in format and will therefore have to be treated accordingly. The workflow described by the flowchart above will guide the post-mission data cleaning process. The aim of this process is the removal of all corrupted or clearly erroneous data, and the formatting of clean data packs for each team member to use.

Most of the data processing will be conducted onboard during the mission campaign, applying calibration methods to raw data (MLX90640, MS5611) and converting all results into easily transmittable formats.

6 Data Storage

The reliable storage of all relevant readings and measurements will be handled by the Raspberry Pi Pico. A script for logging data will be stored on the Pico prior to launch. The data will be directed onto an SD card of size 32 GB. The data storage policy of the mission will address the sequencing of data storage. Sequencing is necessary to avoid conflicts between sensors, which can result in corrupted or missing information.

6.1 Storage Hierarchy

A hierarchical data storage policy will be implemented to manage potential storage or transmission conflicts. This prioritisation ensures that, in the event of limited storage capacity or downlink bandwidth, lower-priority data are sacrificed in favour of mission-critical information.

1. The science data (and any dependent information) will be given first preference: the thermal sensor, valve thermometer, ambient pressure measurements.

2. Telemetry information relating to customer requirements (GPS, ambient temperature). Telemetry data is beneficial for monitoring payload status, but occasional gaps are tolerable as they can be filled via extrapolation.

3. Housekeeping data (battery voltage, error logs).

6.2 Failure Mitigation

A conflict between two of these tiers will delay or skip the lower priority data. The bottom line is that no science data will be overwritten. Within the science tier, priority will be given to the thermal sensor first, and the pressure data second. Due to the intensive calibration process involved in the generation of the thermal array data, it is imperative that backlogs in the data buffer are minimised.

Writing will be staged in a single-file manner, with only one sensor having access to the SD card at any given time. Following from this, a key phase to be considered will be the transitions between file writing. These are the primary stress points of the data storage process. Files will have to be safely closed once data writing is complete to avoid entries from numerous sources. At mission close, the SD card will contain formatted data files for all relevant outputs and flags. A controlled ejection procedure will ensure that all data remains uncorrupted and ready for processing.

References

[1] Melexis. *MLX90640 32x24 IR Array Datasheet*, 2019. Revision 12.